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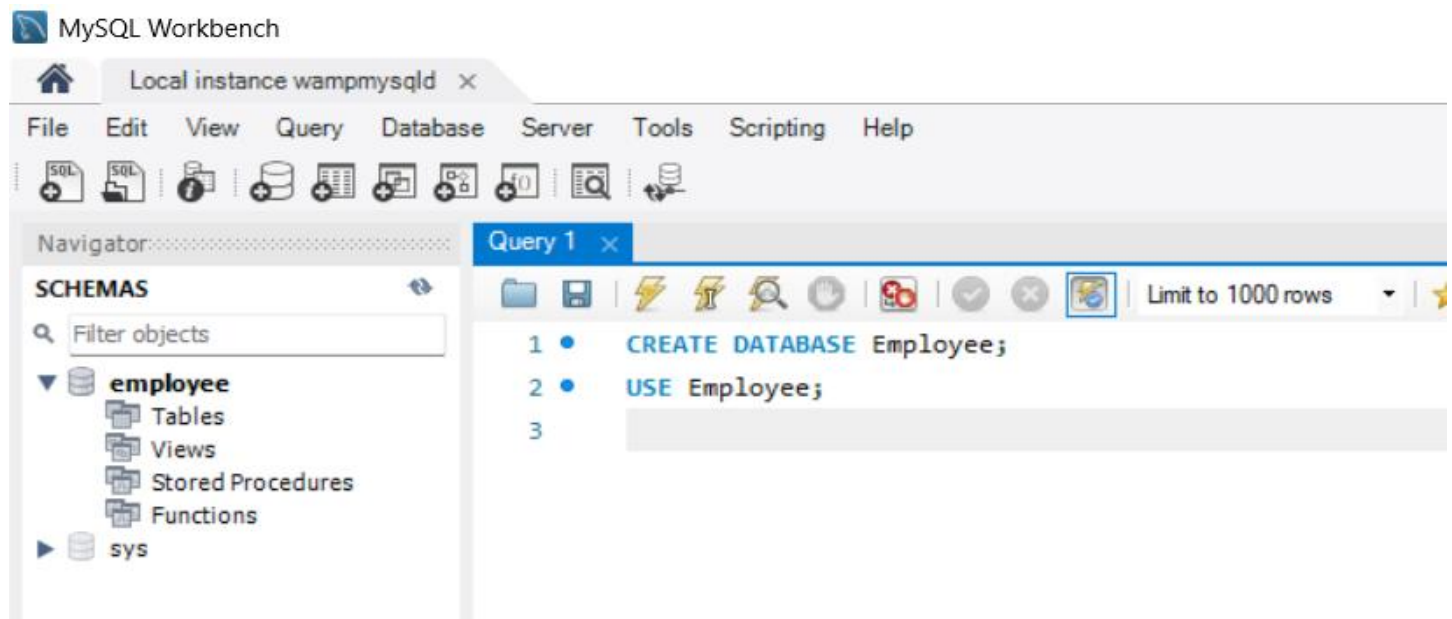
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MySQL Assignment 1 – DDL Commands & Constraints

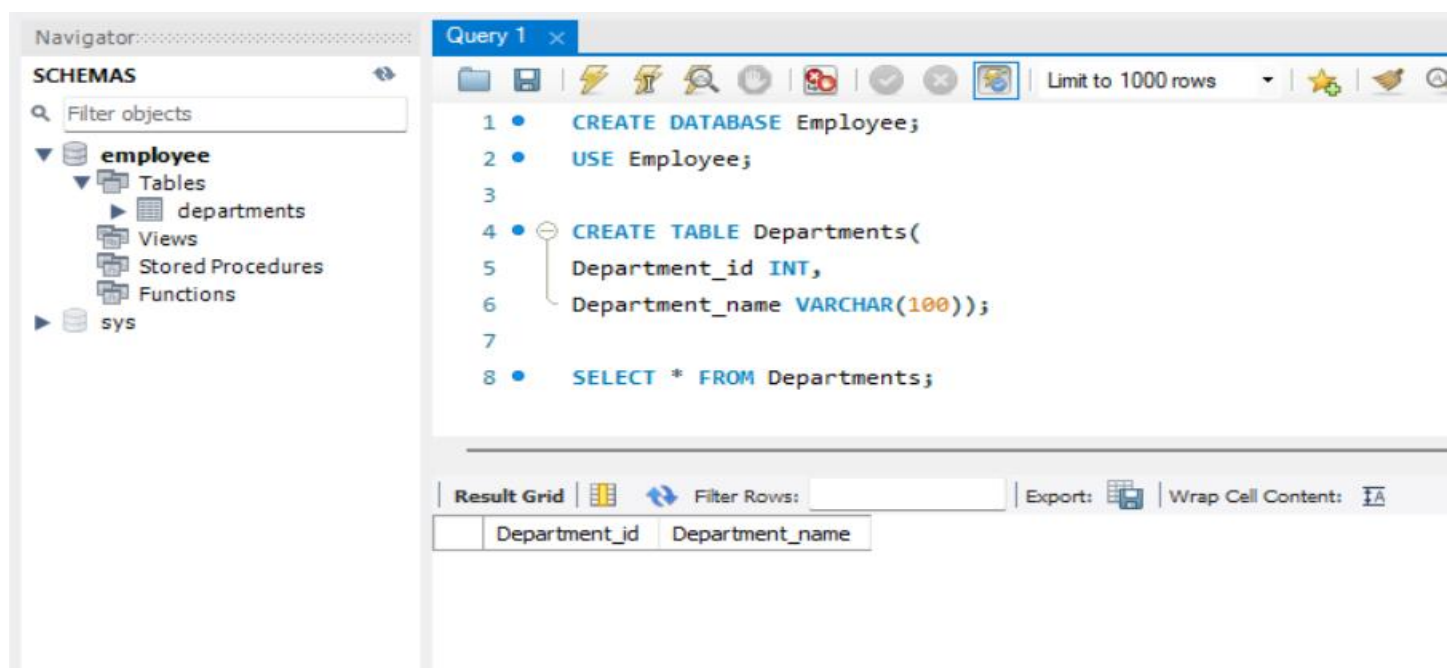
DDL Commands:

1. Table Creation (CREATE): Write the SQL statements to create a database named “employee” and the following tables based on the provided schema:

☉ Departments ☉ Location ☉ Employees



Employee Database



Departments Table

Navigator

SCHEMAS

Filter objects

- employee
 - Tables
 - departments
 - employees
 - Views
 - Stored Procedures
 - Functions
- sys

Query 1

Limit to 1000 rows

```

9
10 CREATE TABLE Employees(
11     Employee_id INT,
12     Employee_name VARCHAR(50),
13     Gender ENUM('M','F'),
14     Age INT,
15     Hire_date DATE,
16     Designation VARCHAR(100),
17     Department_id INT,
18     Location_id INT,
19     Salary DECIMAL(10,2));
20
21 SELECT * FROM Employees;

```

Result Grid

Employee_id	Employee_name	Gender	Age	Hire_date	Designation	Department_id	Location_id	Salary
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Employees Table

Navigator

SCHEMAS

Filter objects

- employee
 - Tables
 - departments
 - employees
 - location
 - Views
 - Stored Procedures
 - Functions
- sys

Query 1

Limit to 1000 rows

```

22
23 CREATE TABLE Location(
24     Location_id INT,
25     Location VARCHAR(30));
26
27 SELECT * FROM Location;

```

Result Grid

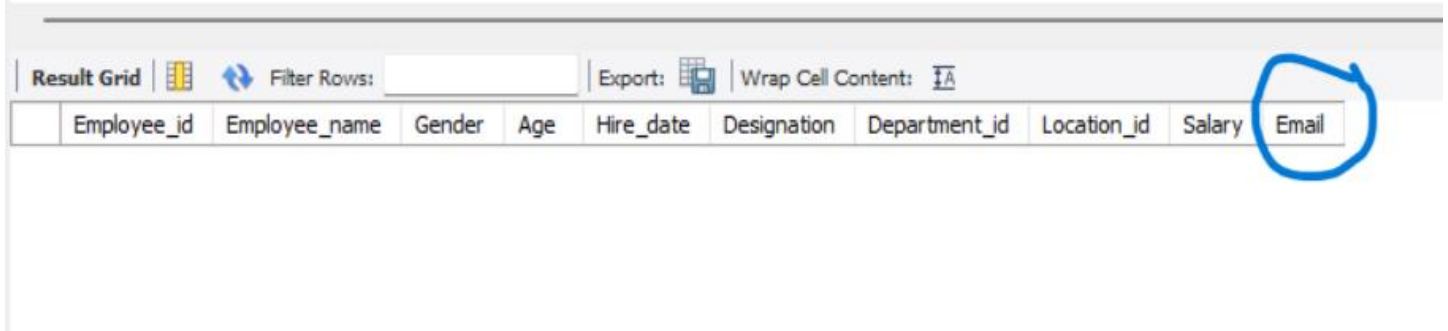
Location_id	Location
-------------	----------

Location Table

2. Table Alteration (ALTER): Consider the following scenarios and write the SQL statements to alter the structure of the tables accordingly:

a) Add a new column named "email" to the Employees table to store employee email addresses.

```
28
29 • ALTER TABLE Employees ADD COLUMN Email VARCHAR(50);
30
```

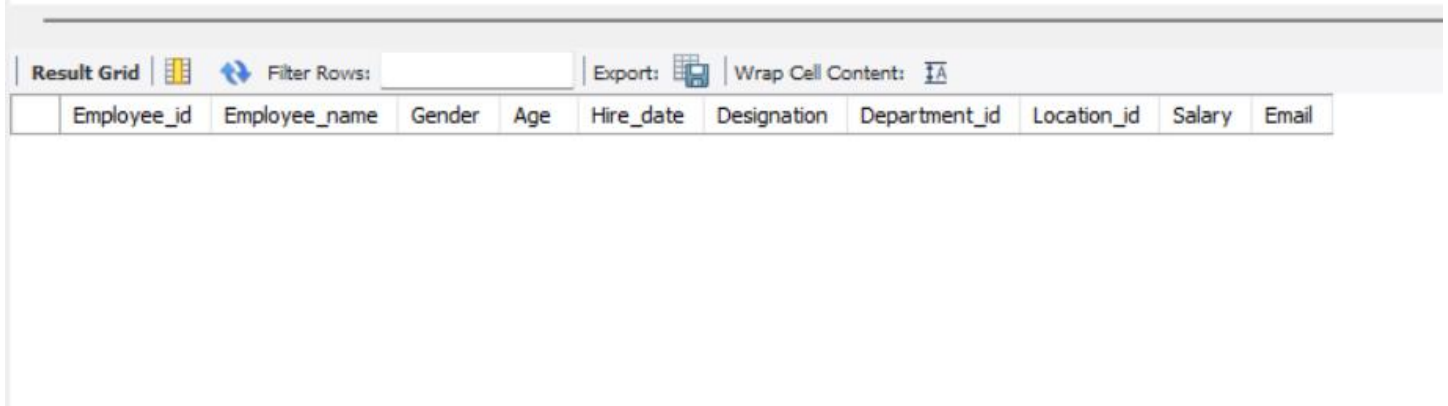


Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Employee_id	Employee_name	Gender	Age	Hire_date	Designation	Department_id	Location_id	Salary	Email
-------------	---------------	--------	-----	-----------	-------------	---------------	-------------	--------	-------

b) Modify the data type of the "designation" column in the Employees table to support a wider range of values.

```
30
31 • ALTER TABLE Employees MODIFY COLUMN Designation VARCHAR(200);
32
```



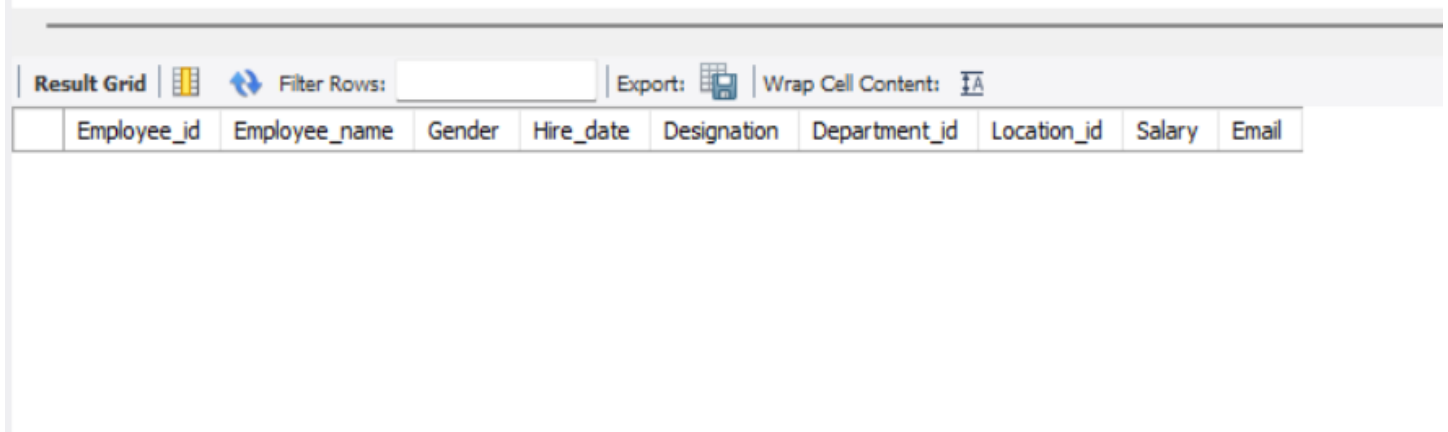
Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Employee_id	Employee_name	Gender	Age	Hire_date	Designation	Department_id	Location_id	Salary	Email
-------------	---------------	--------	-----	-----------	-------------	---------------	-------------	--------	-------

The designation column was modified from VARCHAR(100) to VARCHAR(200)

c) Drop the "age" column from the Employees table.

```
32
33 • ALTER TABLE Employees DROP COLUMN Age;
34
```

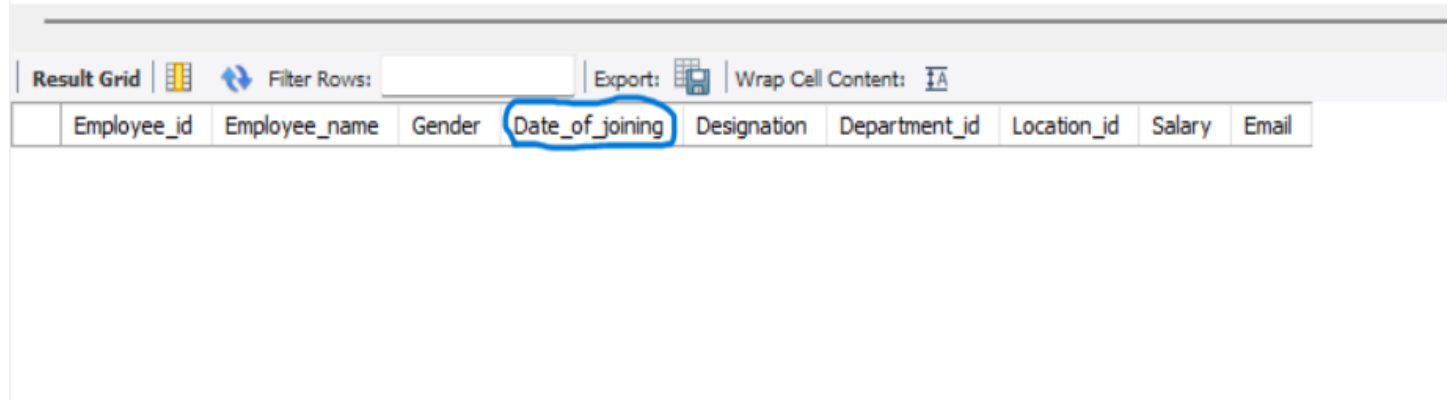


Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Employee_id	Employee_name	Gender	Hire_date	Designation	Department_id	Location_id	Salary	Email
-------------	---------------	--------	-----------	-------------	---------------	-------------	--------	-------

d) Rename the "hire_date" column to "date_of_joining".

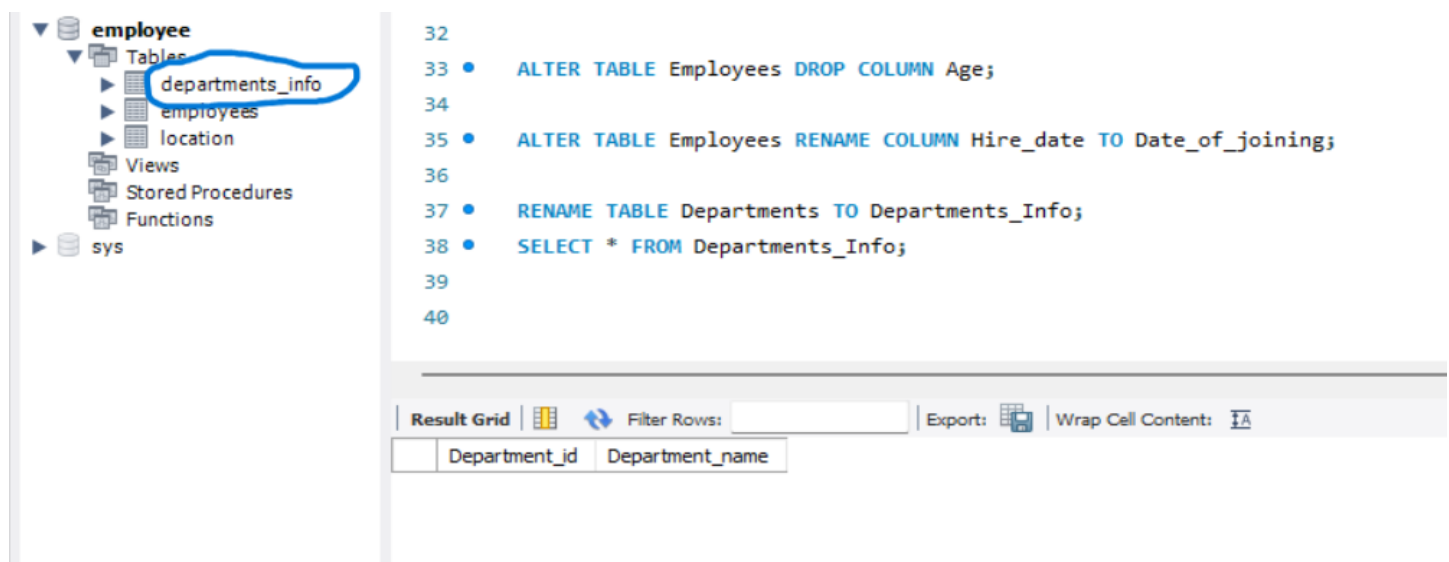
```
34
35 • ALTER TABLE Employees RENAME COLUMN Hire_date TO Date_of_joining;
36
```



The screenshot shows the SQL Developer interface. The top toolbar includes 'Result Grid', 'Filter Rows', 'Export', and 'Wrap Cell Content'. Below the toolbar is a table with the following columns: Employee_id, Employee_name, Gender, Date_of_joining, Designation, Department_id, Location_id, Salary, and Email. The 'Date_of_joining' column is highlighted with a blue circle.

3. Table Renaming (RENAME): Rewrite the SQL statements to rename the following tables:

a) Rename the "Departments" table to "Departments_Info".

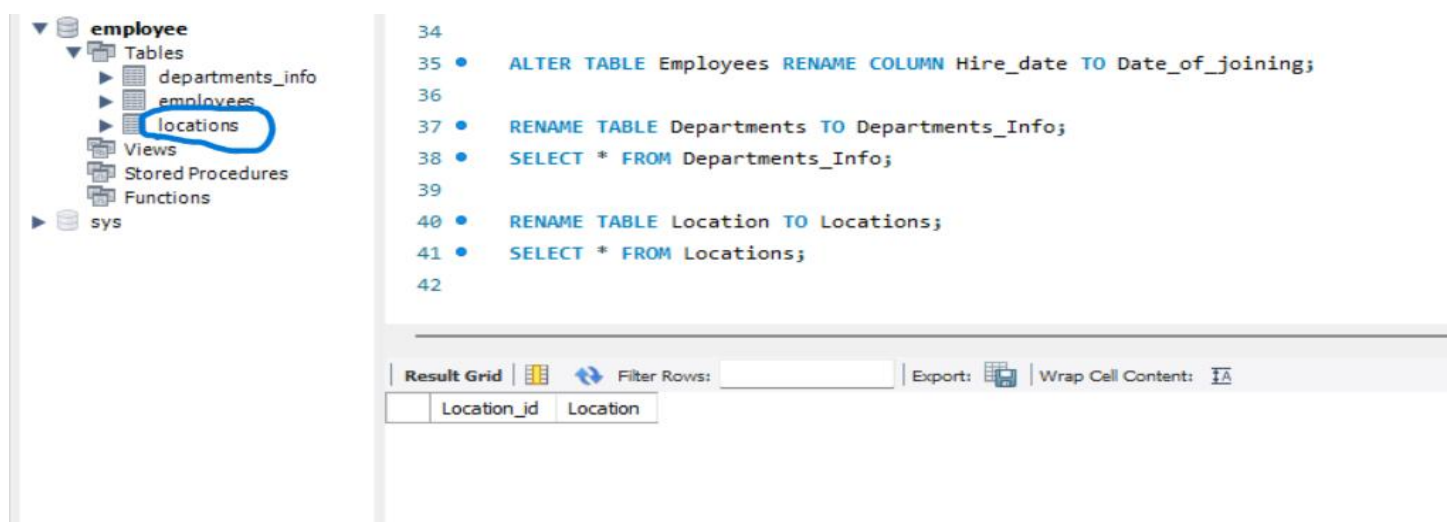


The screenshot shows the SQL Developer interface. On the left, the 'employee' schema is expanded, and the 'departments_info' table is selected. The main pane displays the following SQL statements:

```
32
33 • ALTER TABLE Employees DROP COLUMN Age;
34
35 • ALTER TABLE Employees RENAME COLUMN Hire_date TO Date_of_joining;
36
37 • RENAME TABLE Departments TO Departments_Info;
38 • SELECT * FROM Departments_Info;
39
40
```

Below the SQL statements is a table with the following columns: Department_id and Department_name.

b) Rename the "Location" table to "Locations".

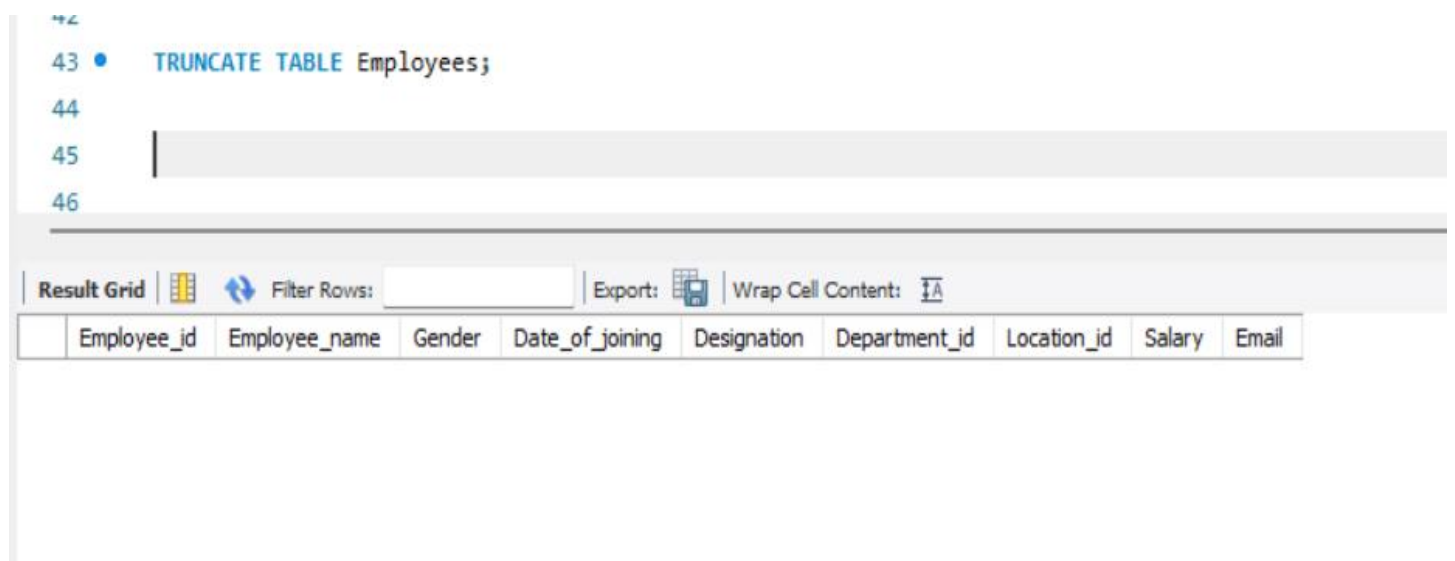


The screenshot shows the SQL Developer interface. On the left, the 'employee' schema is expanded, and the 'locations' table is selected. The main pane displays the following SQL statements:

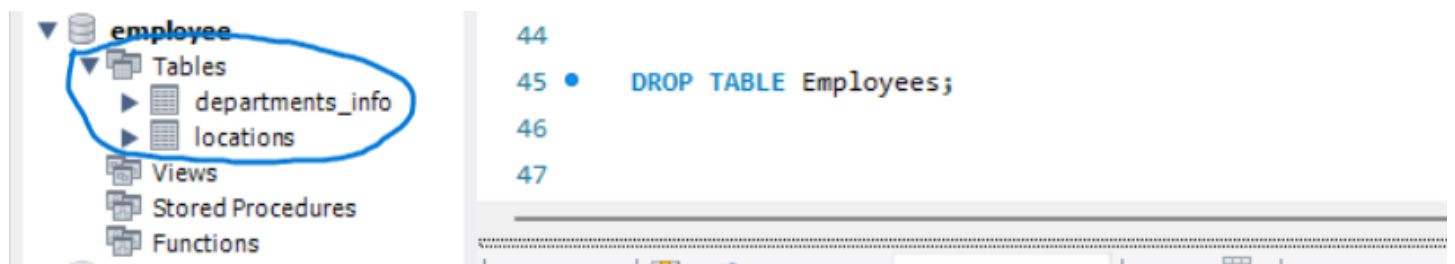
```
34
35 • ALTER TABLE Employees RENAME COLUMN Hire_date TO Date_of_joining;
36
37 • RENAME TABLE Departments TO Departments_Info;
38 • SELECT * FROM Departments_Info;
39
40 • RENAME TABLE Location TO Locations;
41 • SELECT * FROM Locations;
42
```

Below the SQL statements is a table with the following columns: Location_id and Location.

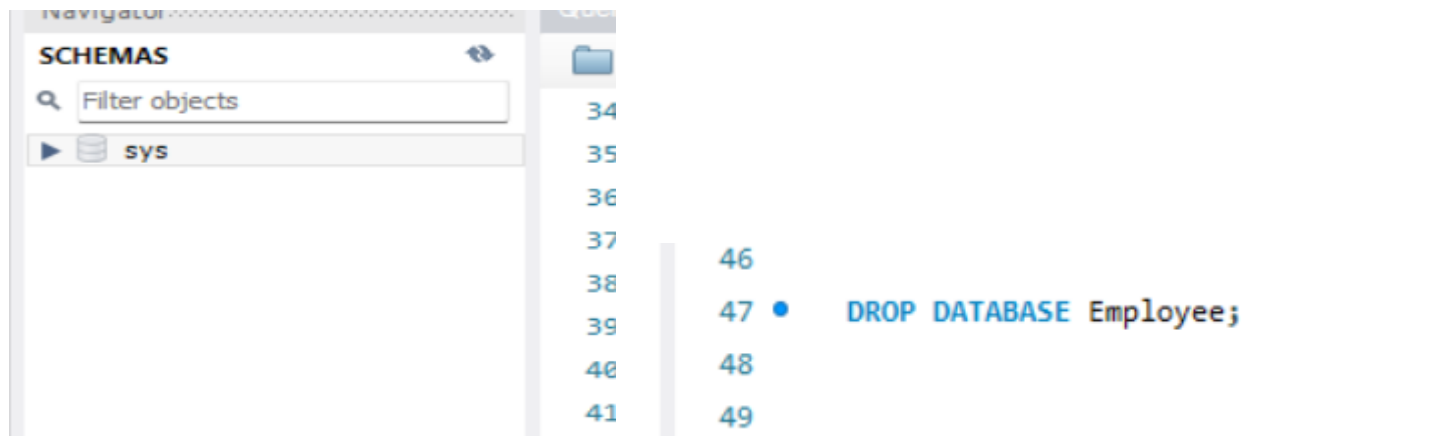
4. Table Truncation (TRUNCATE): Write an SQL statement to truncate the Employees table.



5. Database & Table Dropping (DROP): Write the SQL statements to drop the Employees table and then the “employee” database.



The Employees table was dropped successfully.

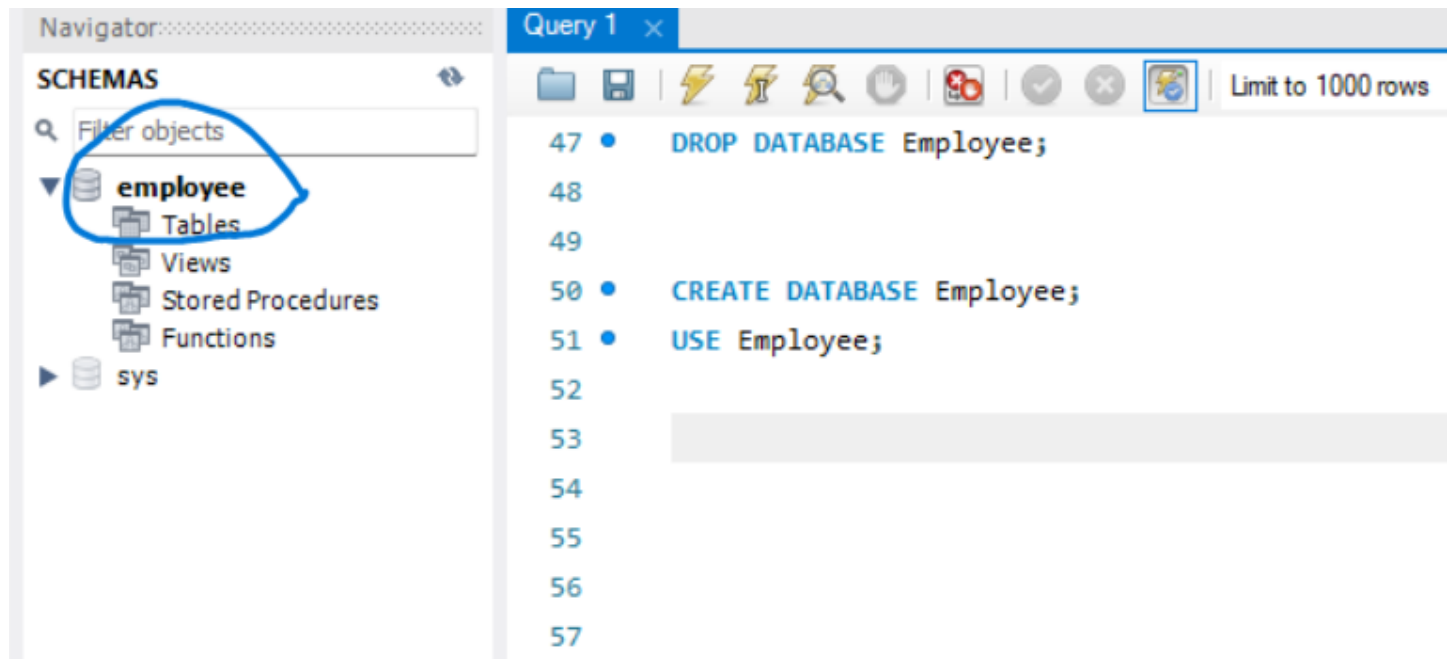


The employee database was dropped successfully.

Constraints:

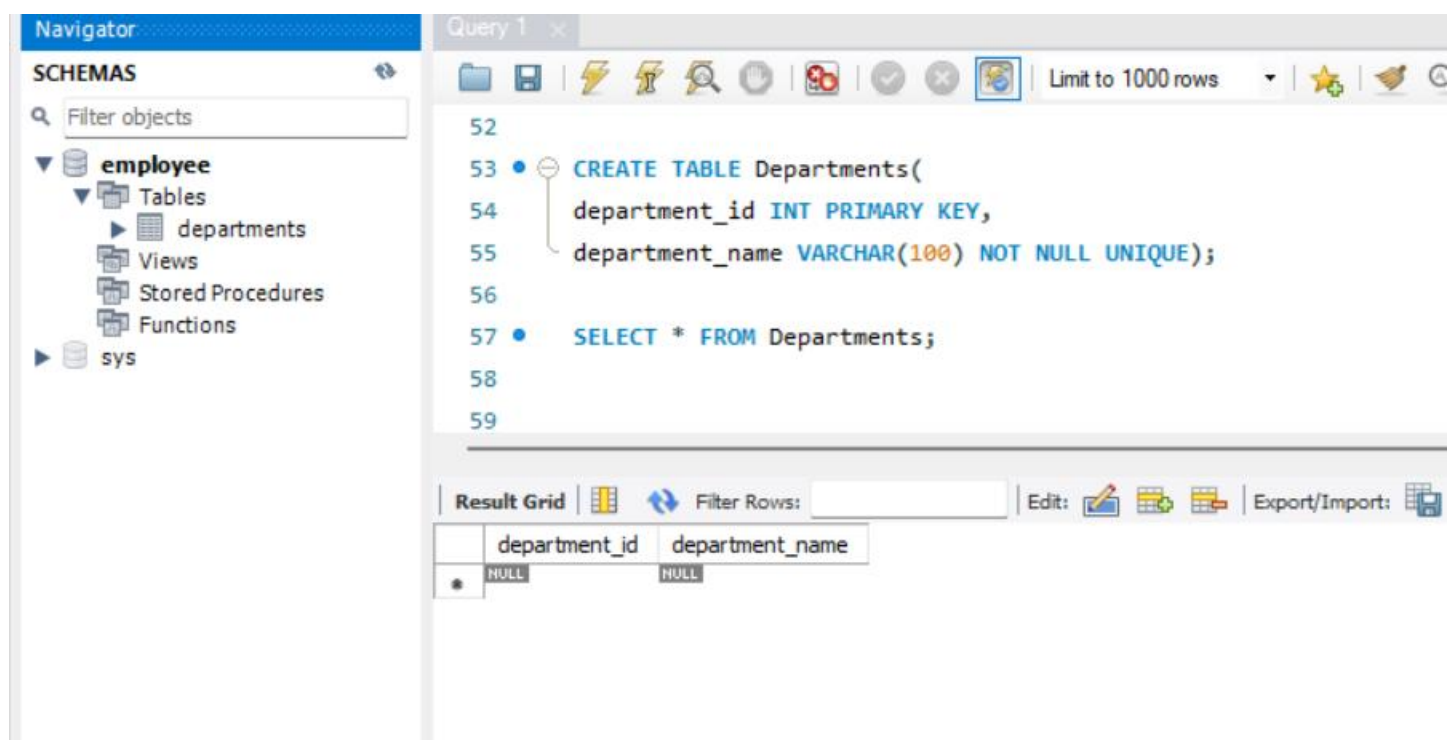
1. Database Recreation:

Drop the 'employee' database if it exists and recreate it using the provided schema, ensuring that all tables are created with the appropriate constraints as instructed.



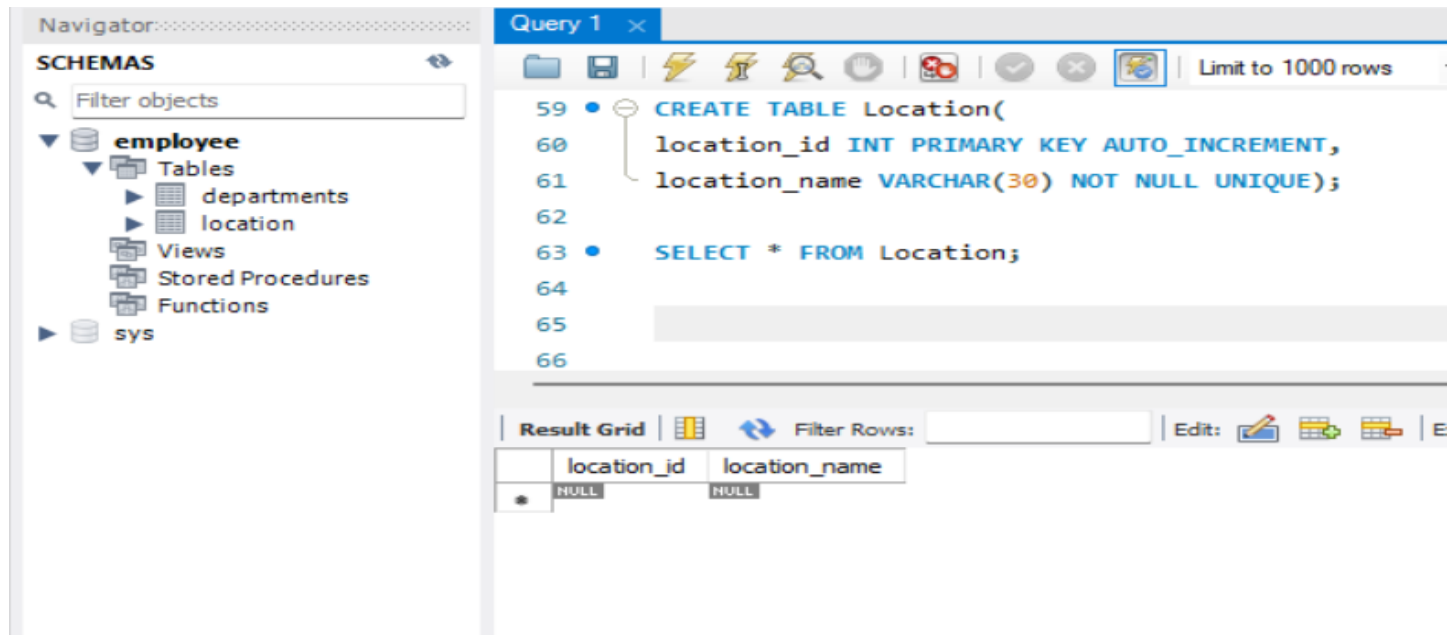
2. Departments Table:

- Ensure that the "department_id" uniquely identifies each department.
- Set up constraints on the "department_name" to avoid duplicate and null entries.



3. Location Table:

- Establish a mechanism to automatically generate unique identifiers for each location, ensuring that they are incremented sequentially.
- Implement constraints to prevent the insertion of null and duplicate locations.



4. Employees Table:

- Guarantee that each employee has a distinct identifier.
- Create a restriction to ensure that the employee's name is always provided.
- Limit the acceptable values for the "gender" field to only 'M' or 'F'.
- Enforce a condition to ensure that the employee's age is 18 or above.
- Automatically assign the current date to the "hire_date" field if not specified.
- Establish links between the "department_id" and "location_id" fields in the "employees" table and their respective tables.

